

Task 6 — Insights & Business Recommendations

Customer Churn Prediction & Risk Segmentation Dashboard

1. Which model performed best and why did you select it?

Among the 3 models, Logistic Regression achieved higher metric scores on ROC-AUC, Accuracy, and F1-Score. However, when considered at a larger scale of data, Logistic Regression may underperform compared to XGBoost. Additionally, features like Tenure and Contract Type exhibit more of a linear dependency, which naturally favours Logistic Regression on this dataset.

In short-scale prediction, Logistic Regression is a good choice. For large-scale, production-level prediction, XGBoost is a better model due to its ability to capture non-linear interactions, handle feature interactions automatically, and generalise well across diverse customer segments.

2. What are the top 3 factors driving customer churn?

Contract Type

Based on the type of contract taken, customers on month-to-month plans churn at a much higher rate (43%) compared to those on a one-year plan (11%) or a two-year plan (3%). Offering a 3-month plan could serve as a middle ground for customers who are unable to commit to a yearly plan.

Tenure

First impressions matter. Customers who churn from the monthly plan do so largely due to early disappointment with the service or product quality. Customers who have maintained longer tenures tend to stay loyal. This highlights the critical importance of strong onboarding and early service quality in the first few months of a customer's lifecycle.

Monthly Charges

High-charge customers with no long-term contracts are far more likely to churn, especially when paired with low tenure and a lack of value-added services such as TechSupport or OnlineBackup. These customers perceive low value relative to cost and are prime candidates for proactive retention.

3. What do High Risk customers have in common?

High-risk customers (churn probability ≥ 0.70) are predominantly on month-to-month contracts, have short tenures typically under 12 months, and carry above-average monthly charges. They often lack value-added services like TechSupport or OnlineSecurity, and are more likely to be senior citizens without any support services enrolled.

4. Two Actionable Business Recommendations

Early Loyalty Program

Automatically enroll new customers with high churn probability scores into a targeted loyalty program. Offer a 10–15% discount or a free service upgrade (such as

TechSupport or OnlineBackup) at the 3-month mark if they upgrade to a one-year contract. This directly addresses the two biggest churn drivers — short tenure and month-to-month contracts — simultaneously.

Personalised Care for High-Risk Segment

Run a monthly batch of the churn model on the active customer base. Flag all customers in the High Risk tier and assign them to a dedicated retention team. Personalise outreach based on each customer's specific gap: if they lack TechSupport, offer a 30-day free trial; if they have high charges, present a comparable plan at a lower price. Proactive retention is 5× cheaper than re-acquiring lost customers.

5. Limitations & Potential Improvements

- The model does not account for temporal patterns. Customer behaviour changes over time, and a time-series approach such as survival analysis could better capture churn timing.
- No customer satisfaction or NPS data is available. Including support ticket volume or satisfaction scores would significantly improve recall on high-risk customers.
- The dataset is from a single telecom provider and may not generalise to other industries or geographies.
- With more time, SHAP (SHapley Additive exPlanations) values would allow per-customer explainability, which is essential for a real-world retention team to act on model outputs confidently.